Temi Otun

otun@ualberta.ca | GitHub | LinkedIn | Website

Education

University of Alberta — Edmonton, AB

Bachelor of Science, Computing Science – Artificial Intelligence Specialization, Minor in Business — *Sep. 2022 – Present*

- **Awards:** Jason Lang Scholarship, Dean's List

Experience

Data Engineering Intern — May 2026 – Aug. 2026

Royal Bank of Canada (RBC) – Amplify Program — *Vancouver, BC*

- Developed a **patent-pending fraud detection system** combining ML classification with LLM analysis, achieving **94% recall** and an **85% F1-score**, projected to mitigate **$18M in annual fraud losses**
- Engineered a cascaded NLP pipeline processing **23M+ annual unstructured text records** using semantic embeddings and multi-stage classification, reducing operational costs by **65%** (**$1.5M in projected annual savings**)
- Led model experimentation across **6+ ML architectures**, optimizing ensemble thresholds, improving precision from **49% to 77%** and supporting projected protection of **12K+ clients annually**

Machine Learning Research Intern — Jan. 2025 – Aug. 2025

The Metabolomics Innovation Centre — *Edmonton, AB*

- Enhanced long-range temperature forecasting system, resulting in up to **47%** lower MAE with an average **24.3%** improvement across a six-year test set compared to previous approaches
- Built multivariate forecasting pipelines across **100+** meteorological features using feature selection, temporal validation, and ensemble ML, outperforming baselines in **73%** of test years
- Assembled forecasting pipelines with advanced feature engineering and benchmarked **20+** ML models, attaining a **42%** accuracy improvement over baseline methods

Machine Learning Research Assistant — Sept. 2024 – Apr. 2026

University of Alberta — *Edmonton, AB*

- Developed a **3-class** dementia and mild cognitive impairment detection system using Whisper-derived speech embeddings and ensemble classifiers, contributing to research presented at **ICASSP 2025**
- Built survival-analysis and clinical prediction pipelines on MIMIC-IV, processing **59M+ patient-event records** across a cohort of **239K+ patients** using temporal preprocessing and feature engineering
- Designed and evaluated **15+** classification and regression experiments across **3 data modalities**: speech, ECG, and clinical records, using hyperparameter optimization and statistical performance analysis

Data Management Intern — Jan. 2024 – May 2024

InfoStrux — *Vancouver, BC*

- Optimized **25+** SQL queries in Snowflake, reducing execution time by up to **60%** and boosting performance of business intelligence dashboards
- Partnered with a senior data engineer to design Snowflake staging and curated layers for **3** datasets; wrote and tuned **30+** queries, improving data reusability and cutting time-to-insight by **40%**

Projects

Local LLM Agent System | *LLMs, RAG, AI Agents, MCP, REST APIs, Information Retrieval*

- Engineered a local multimodal RAG pipeline with **Qwen 3.5**, **Ollama**, and **Docling**, combining **Nomic embeddings**, hybrid retrieval, and **BGE reranking** for document indexing
- Indexed **100+** lecture documents for source-grounded Q&A with inline citations, improving top-5 retrieval recall from **64% to 86%** through BGE cross-encoder reranking
- Built an **MCP**-based Gmail agent on the same self-hosted LLM, automating **200+** outreach and follow-up emails for student-club recruiting and partnership outreach

Event Lottery System | *Java, Kotlin, Firebase, Gemini API, Google Maps* — GitHub

- Collaborated in a **seven-person Agile team** to build a Firestore-backed Android platform supporting **3 user roles** and end-to-end lottery workflows, including waitlists, randomized selection, invitations, and automatic redraws
- Integrated QR-code registration, geolocation, and real-time notifications with a serverless **Gemini-powered classification pipeline** that mapped user-generated event tags across **21 validated categories**

Technical Skills

Programming: Python, SQL, Java, C
ML/AI: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, RAG, AI Agents, Time Series, Reinforcement Learning, Feature Engineering, Model Evaluation
Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, XGBoost, LightGBM, CatBoost, Django
Data & Infrastructure: PostgreSQL, Snowflake, Firebase, Docker, Kubernetes, Linux, AWS S3, Git
AI Systems: MCP, REST APIs, Vector Search, Embeddings, Semantic Search, Local LLM Inference